PRACTICAL 7:

Practical7(1)

Problem Phase:

A government office issues tokens to visitors in the order they arrive. The queue has a fixed capacity of n tokens. New tokens are added at the rear, and tokens are served from the front. If the queue is full, a new token cannot be added. If the queue is empty, no token can be served. The program prints the front token after each operation and reports errors for invalid operations.

Solution:

The front pointer points to the first token, and the rear pointer points to the last token. Insertion moves rear forward, and serving moves front forward. A linear queue cannot reuse freed slots at the beginning.

Practical7(2)

Problem Phase:

A hospital emergency ward attends to patients in arrival order. New patients join at the rear, and the doctor attends to the patient at the front. The queue has no predefined capacity. The program prints the front patient after each operation and reports an error if an attend operation is attempted on an empty queue.

Solution:

A linked list queue uses dynamic memory, so it has no fixed capacity. The front patient is attended first, and if the queue is empty, the program displays an error message.